QF1100 Introduction to Quantitative Finance — Final Exam Cheatsheet

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1. Theory of Interest

Accumulation Functions

Simple: $a(t) = 1 + rt$ **Compound:** $a(t) = (1+r)^t$ [default]

Continuous: $a(t) = e^{\delta t}$

Accumulated value of principal P: $A(t) = P \cdot a(t)$

Frequency of Compounding

Nominal rate $r^{(p)}$ paid p times/yr:

$$a(t) = \left(1 + \frac{r^{(p)}}{p}\right)^{pt}$$

Effective annual rate:

$$r_e = \left(1 + \frac{r^{(p)}}{p}\right)^{p} - 1$$

Equivalence of two nominal rates:

$$\left(1 + \frac{r^{(p)}}{p}\right)^{p} = \left(1 + \frac{R^{(q)}}{q}\right)^{q}$$

$r_e > r^{(p)}$ for any $p > 1$. Continuous compounding maximises r_e for a given nominal rate.

Force of Interest

$$\delta(t) = \frac{a'(t)}{a(t)} = \frac{d}{dt}\ln a(t)$$

Constant δ: $a(t) = e^{\delta t}$, $\delta = \ln(1+r_e)$, $r_e = e^{\delta} - 1$.

Variable force of interest:

$$a(t) = \exp\Big(\int_0^t \delta(u)\,du\Big) \quad a(s,t) = \exp\Big(\int_s^t \delta(u)\,du\Big)$$

Key Conversions (quick ref)

Convert	Formula
$r^{(p)} \to r_e$	$(1 + r^{(p)}/p)^p - 1$
$r_e \to r^{(p)}$	$p[(1+r_e)^{1/p} - 1]$
$r_e \to \delta$	$\ln(1+r_e)$
$\delta \to r_e$	$e^{\delta} - 1$
$r^{(p)} \to r^{(q)}$	equate $(1 + r^{(p)}/p)^p = (1 + r^{(q)}/q)^q$

2. Present Value & Cash Flows

Present Value (single payment)

$$\mathrm{PV} = \frac{c}{a(t)} = \frac{c}{(1+r)^t}$$

Discount factor: $v = \frac{1}{1+r}$, so $\mathrm{PV} = c \cdot v^t$

PV of Cash Flow Stream

$C = \{(c_i, t_i)\}_{i=1}^n$:

$$\mathrm{PV}(C) = \sum_{i=1}^{n} \frac{c_i}{(1+r)^{t_i}}$$

Time value at t: $\mathrm{TV}_t(C) = \mathrm{PV}(C) \cdot (1+r)^t$

Annuities

Let r = effective rate per period, A = payment per period, n = number of periods.

Ordinary annuity (payment at *end* of each period):

$$\mathrm{PV} = \frac{A}{r}\left[1 - \frac{1}{(1+r)^n}\right]$$

Annuity-due (payment at *start* of each period):

$$\mathrm{PV} = \frac{A(1+r)}{r}\left[1 - \frac{1}{(1+r)^n}\right]$$

$\Rightarrow$ PV(due) = PV(ordinary) $\times (1+r)$

Perpetuity (ordinary, payments at end):

$$\mathrm{PV} = \frac{A}{r}$$

Perpetuity-due (payments at start):

$$\mathrm{PV} = \frac{A(1+r)}{r}$$

Future value (ordinary annuity at $t = n$):

$$\mathrm{FV} = \frac{A}{r}[(1+r)^n - 1]$$

Deferred annuity (n payments, first at $t = m+1$):

$$\mathrm{PV} = \frac{A}{r}\left[1 - \frac{1}{(1+r)^n}\right] \cdot \frac{1}{(1+r)^m}$$

Discount ordinary annuity PV back by m periods.

Rate mismatch: Always convert nominal rate to *effective rate per payment period* before applying annuity formula. E.g. 6% p.a. convertible semi-annually $\to$ effective monthly rate $= (1+0.03)^{1/6} - 1$.

NPV & IRR Decision Rules

NPV criterion: Invest if NPV > 0. Prefer A over B if $\mathrm{NPV}(A) > \mathrm{NPV}(B)$.

IRR criterion: Invest if IRR $>$ required return. Prefer A over B if $\mathrm{IRR}(A) > \mathrm{IRR}(B)$.

IRR is r s.t. $\mathrm{PV}(C) = 0$. **Uniqueness:** if $c_0 < 0$, $c_k \ge 0\ \forall k \ge 1$, $\sum c_k > 0 \Rightarrow$ unique IRR.

Newton–Raphson: $r_{n+1} = r_n - f(r_n)/f'(r_n)$

Growing Annuity & Perpetuity

First payment A at $t = 1$, grows at rate g/period, n periods:

$$\mathrm{PV} = \frac{A}{r-g}\left[1 - \left(\frac{1+g}{1+r}\right)^n\right] \quad (r \ne g) \qquad r = g: \frac{nA}{1+r}$$

Growing perpetuity ($r > g$): $\mathrm{PV} = A/(r-g)$.

Loans, Refinancing & Last Payment

Loan L, n payments A, rate r/period: $L = \frac{A}{r}[1 - (1+r)^{-n}]$, $A = \frac{Lr}{1-(1+r)^{-n}}$

Balance after m payments: $L_m^{\text{bal}} = \frac{A}{r}[1 - (1+r)^{-(n-m)}]$ (prospective)

$L_m^{\text{bal}} = L(1+r)^m - \frac{A}{r}[(1+r)^m - 1]$ (retrospective)

Irregular last payment (n non-integer): make $\lfloor n \rfloor$ full payments of A plus final smaller payment B.

- B at $t = \lceil n \rceil$: set L = PV of $\lfloor n \rfloor - 1$ payments of A plus B at $t = \lceil n \rceil$.
- B at $t = \lfloor n \rfloor + 1$: use $L_{\lfloor n \rfloor}^{\text{bal}}$, then $B = L_{\lfloor n \rfloor}^{\text{bal}}(1+r)$.

3. Bonds

Notation

F	face value ($= R$ unless stated)
R	redemption value
c	nominal coupon rate (annual)
m	coupon payments per year
n	total coupon payments
λ	nominal bond yield (annual)
P	bond price

Bond Pricing Formula

$$P = \frac{R}{(1+\lambda/m)^n} + \sum_{i=1}^{n} \frac{cF/m}{(1+\lambda/m)^i}$$

$$= F\left[\frac{1}{(1+\lambda/m)^n} + \frac{c/m}{\lambda/m}\left(1 - \frac{1}{(1+\lambda/m)^n}\right)\right] \quad (F = R)$$

Premium/discount form ($F = R$): $P = F + F \cdot \frac{c-\lambda}{\lambda}[1 - (1+\lambda/m)^{-n}]$

$c > \lambda \Rightarrow P > F$ (premium); $c = \lambda \Rightarrow P = F$ (par); $c < \lambda \Rightarrow P < F$ (discount).

Makeham formula (alternative; let $K = R/(1+\lambda/m)^n$):

$$P = K + \frac{c}{\lambda}(F - K) \qquad (F = R \text{ case})$$

Recursive price: $P_{k+1} = P_k(1+\lambda/m) - cF/m$ (price after k-th coupon).

Variable Coupon Bonds

Split into groups with constant coupon rate; price each group as a separate annuity + shared redemption PV.

Exam pattern (Q3 midterm): Bond with different coupon rates for each 5-year block and $R \ne F$. Price each block as an annuity discounted at yield λ, then add PV of redemption at maturity.

Determining Bond Yield (Newton–Raphson)

$f(\lambda) = P(\lambda) - P_{\text{market}}$, iterate $\lambda_{n+1} = \lambda_n - f(\lambda_n)/f'(\lambda_n)$.

$$P(\lambda) = \frac{R}{(1+\lambda/m)^n} + \frac{cF/m}{\lambda/m}\left[1 - \frac{1}{(1+\lambda/m)^n}\right]$$

$f'(\lambda)$ for annual coupon bond ($m = 1$, $F = R$):

$$f'(\lambda) = -\sum_{i=1}^{n} \frac{i \cdot cF}{(1+\lambda)^{i+1}} - \frac{nF}{(1+\lambda)^{n+1}} = -\frac{D \cdot P(\lambda)}{1+\lambda}$$

where D = Macaulay duration. (In general: $f'(\lambda) = -\frac{D \cdot P}{m(1+\lambda/m)}$.)

Start: $\lambda_0 = c$. Price–yield: **inverse** & **convex** ($P \downarrow$ as $\lambda \uparrow$).

Macaulay Duration

$$D = \frac{\sum_{i=1}^n t_i \cdot \mathrm{PV}(c_i, t_i)}{\mathrm{PV}(C)} = \sum_{i=1}^{n} w_i t_i, \quad w_i = \frac{\mathrm{PV}(c_i)}{\mathrm{PV}(C)}$$

Special cases ($\mu = \lambda/m$, $\gamma = c/m$): Zero-coupon: $D = T$. Perpetuity ($m = 1$): $D = (1+\lambda)/\lambda$.

Coupon bond, $F = R$ (general): $$D = \frac{1+\mu}{m\mu} - \frac{1 + \mu + n(\gamma - \mu)}{m\gamma[(1+\mu)^n - 1] + m\mu} \quad \textbf{(2)}$$

At par $P = F = R$ ($\mu = \gamma$): $$D = \frac{1+\mu}{m\mu}\left[1 - \frac{1}{(1+\mu)^n}\right] \quad \textbf{(3)}$$

$t_1 \le D \le t_n$; D independent of F and of yield compounding frequency.

4. Term Structure

Spot Rates & Forward Rates

Spot rate s_t: effective annual rate from 0 to t. Accumulation: $(1+s_t)^t$.

Forward rate $f_{j,k}$: rate from $t = j$ to $t = k$ (annual, same convention as spot).

No-arbitrage relation:

$$(1+s_k)^k = (1+s_j)^j (1+f_{j,k})^{k-j}$$

Note: $f_{0,t} = s_t$.

Continuous compounding version:

$$e^{s_k k} = e^{s_j j} \cdot e^{f_{j,k}(k-j)} \quad \Rightarrow \quad f_{j,k} = \frac{s_k k - s_j j}{k - j}$$

Bootstrapping Spot Rates

Use observed coupon bond prices to extract $s_1, s_2, \ldots$ sequentially:

For a coupon bond with annual payments:

$$P = \frac{c_1}{1+s_1} + \frac{c_2}{(1+s_2)^2} + \cdots + \frac{c_n + F}{(1+s_n)^n}$$

Solve for the unknown s_n given $s_1, \ldots, s_{n-1}$ already found.

Exam pattern: Extract s_1, s_2 from zero-coupon bond prices, then use coupon bond price to find s_3, then compute $f_{1,3}$.

Yield Curve Shapes

Normal (upward): economic expansion, low inflation.

Inverted (downward): pre-recession, high inflation.

Flat term structure: all s_t equal $\Rightarrow$ force of interest is constant.

5. Asset Returns & Portfolios

Returns

Total return: $R = P_1/P_0$ **Rate of return:** $r = (P_1 - P_0)/P_0$ $R = 1 + r$

Short sell: $R_{\text{short}} = P_1/P_0$ (same formula – negatives cancel).

Random Returns & Key Identities

Mean: $\mu_i = E(r_i)$ Variance: $\sigma_i^2 = \mathrm{Var}(r_i)$

Covariance: $\sigma_{ij} = \mathrm{Cov}(r_i, r_j)$ Correlation: $\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$

Var/Cov identities:

- $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$
- $\mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X, Y)$
- $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$
- $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\mathrm{Cov}(X, Y)$

Short-sell gain (sell at P_0, buy back at P_1): profit $= P_0 - P_1$. Rate of return $= (P_0 - P_1)/P_0 = 1 - P_1/P_0 = -r_{\text{long}}$. **Loss is unlimited.**

Portfolio of 2 Assets

Weights $w_1 = \alpha$, $w_2 = 1 - \alpha$ ($\alpha \in \mathbb{R}$ if short-selling allowed; $\alpha \in [0, 1]$ otherwise).

$$\mu_p = \alpha\mu_1 + (1-\alpha)\mu_2$$

$$\sigma_p^2 = \alpha^2\sigma_1^2 + (1-\alpha)^2\sigma_2^2 + 2\alpha(1-\alpha)\rho_{12}\sigma_1\sigma_2$$

Global Minimum-Variance Portfolio (GMVP)

$$\alpha^* = \frac{\sigma_2^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

$$(\sigma_p^2)^* = \frac{\sigma_1^2\sigma_2^2(1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

If $\rho = 1$: feasible set is a straight line. If $\rho = -1$: V-shape (zero variance achievable). If $-1 < \rho < 1$: curved bullet shape.

Efficient frontier: upper half of feasible set (including GMVP).

Check: if $\alpha^* \notin [0,1]$ and short-selling not allowed, the GMVP is the asset with lower σ.

6. Forwards & Futures

Forward Contract

Obligation to buy/sell at forward price $F(t,T)$ at maturity T.

- Long payoff: $S_T - F(t,T)$
- Short payoff: $F(t,T) - S_T$
- Costs nothing to enter (no premium)

Forward Pricing (No-Arbitrage)

Non-dividend-paying asset:

$$F(0,T) = S_0 e^{rT} \text{ (cts)} \quad F(0,T) = S_0(1+r)^T \text{ (disc)}$$

General t: $F(t,T) = S_t\, e^{r(T-t)}$

With storage cost c (lump sum at $t=0$):

$$F(0,T) = (S_0 + c)\, e^{rT}$$

With continuous dividend yield q:

$$F(0,T) = S_0\, e^{(r-q)T}$$

Forward price does **not** depend on expected return μ of the underlying — only on S_0, r, T (and carry costs). This is a no-arbitrage result, not an expectation.

Arbitrage Construction

If $F(0,T) > S_0 e^{rT}$: short forward, buy asset (borrow S_0). Profit at T: $F - S_0 e^{rT} > 0$.
If $F(0,T) < S_0 e^{rT}$: long forward, short-sell asset (invest proceeds). Profit at T: $S_0 e^{rT} - F > 0$.

Futures vs. Forwards

	Forward	Futures
Settlement	At maturity	Daily MTM
Counterparty	Bilateral	Exchange

Daily P&L long futures: $\approx F_{t+1} - F_t$ (margin account).

Margin Account

Initial margin deposited. If balance falls below **maintenance margin**, margin call issued (top up to initial margin).

7. Hedging

Perfect Hedge (Forwards/Futures)

Enter opposite position to fix price and eliminate risk. Conditions for perfect futures hedge: (1) delivery date matches; (2) same underlying; (3) integer multiple of contract size; (4) sufficient liquidity.

Hedging Direction ($\rho > 0$)

Obligation	Futures action
Buy asset A	Short futures B
Sell asset A	Long futures B

If $\rho < 0$: reverse direction.

Minimum-Variance Hedge

Hedge W units of asset A using futures on asset B (correlated). Short h futures contracts.
Cash flow at T: $y = -WS_T + (F_T - F_0)h$

$$\mathrm{Var}(y) = W^2\mathrm{Var}(S_T) - 2W\mathrm{Cov}(S_T, F_T)h + \mathrm{Var}(F_T)h^2$$

Optimal hedge ratio:

$$h^* = W \cdot \frac{\mathrm{Cov}(S_T, F_T)}{\mathrm{Var}(F_T)} = W \cdot \frac{\rho_{S,F}\,\sigma_S}{\sigma_F}$$

Minimum variance:

$$\mathrm{Var}^*(y) = W^2\mathrm{Var}(S_T)(1 - \rho^2_{S,F})$$

Hedge effectiveness: $\rho^2_{S,F}$. Factor $\sqrt{1-\rho^2}$ measures residual risk.

h^* is in *units*, not contracts. Number of contracts = h^*/(contract size). Round to integer for real problems.

8. Options

Key Definitions

Call: right to *buy* at strike K. **Put**: right to *sell* at strike K.
Long = holder (pays premium); Short = writer (receives premium).
European: exercise only at T. American: exercise any time $\le T$. (Assume European unless stated.)

Payoffs & Profits

Position	Payoff at T	Profit
Long call	$\max(S_T - K, 0)$	payoff $-Ce^{rT}$
Short call	$-\max(S_T - K, 0)$	payoff $+Ce^{rT}$
Long put	$\max(K - S_T, 0)$	payoff $-Pe^{rT}$
Short put	$-\max(K - S_T, 0)$	payoff $+Pe^{rT}$

C, P = option premium (time-0 price). Time-value of premium at T: Ce^{rT} or Pe^{rT}.

Bounds

Upper bounds: $C \le S_0$ $\quad P \le Ke^{-rT}$
Lower bounds:

$$C \ge \max(0,\ S_0 - Ke^{-rT}) \qquad P \ge \max(0,\ Ke^{-rT} - S_0)$$

Put–Call Parity (European, no dividends):

$$C + Ke^{-rT} = P + S_0$$

Proof: Portfolio A (long call + zero-coupon bond paying K) and Portfolio C (long put + stock) have identical payoffs ⇒ same price by no-arbitrage.

Trading Strategies

Bull spread: Long call K_1 + Short call K_2, $K_1 < K_2$. Profit if $S_T \uparrow$.
Bear spread: Short call K_1 + Long call K_2 *or* Long put K_2 + Short put K_1. Profit if $S_T \downarrow$.
Butterfly ($K_1 < K_2 < K_3$, $K_1 + K_3 = 2K_2$): Long C_{K_1}, Short $2C_{K_2}$, Long C_{K_3}:

S_T range	Payoff
$S_T \le K_1$	0
$K_1 < S_T \le K_2$	$S_T - K_1$
$K_2 < S_T \le K_3$	$K_3 - S_T$
$S_T > K_3$	0

Profit if $S_T \approx K_2$ (low volatility bet).
Straddle: Long call + Long put (same K, T). Profit if $|S_T - K|$ large.
Covered call: Long stock + Short call. **Protective put:** Long stock + Long put.

Binomial Option Pricing

Single period: $S \to uS$ or dS, risk-free rate r, $R = 1 + r$. No-arb requires $u > R > d$.
Risk-neutral probability: $q = \frac{R-d}{u-d}$, $1 - q = \frac{u-R}{u-d}$

$$C = \frac{1}{R}[q\,C_u + (1-q)\,C_d]$$

$C_u = \max(uS - K, 0)$, $C_d = \max(dS - K, 0)$ for call; $\max(K - \cdot, 0)$ for put.
Multi-period: backward induction. Each node same formula. **CRR:** $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$.

Replication Portfolio (Binomial)

Construct portfolio: x shares + b in risk-free cash. Payoffs:

$$\text{up: } ux + Rb = C_u \qquad \text{down: } dx + Rb = C_d$$

Solve:

$$x = \frac{C_u - C_d}{u-d} \qquad b = \frac{uC_d - dC_u}{R(u-d)}$$

$$C = x + b = \frac{qC_u + (1-q)C_d}{R} \quad \checkmark$$

Arbitrage: if $C > x + b$: short call, long portfolio. If $C < x + b$: long call, short portfolio.

Put–Call Parity (General Form)

Let d = discount factor to maturity. Slides notation:

$$C + dK = P + S_0$$

Discrete: $d = (1+r)^{-T}$. Continuous: $d = e^{-rT}$.
Rearrange to find arbitrage: if $C + dK \neq P + S_0$, one side is cheap.

Black–Scholes (Intro Only)

Stock GBM: $dS = \mu S\,dt + \sigma S\,dz$. CRR → B–S as $\Delta t \to 0$: $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$.

Exam Patterns (Midterm)

Q1 type — PV algebra: Set up PV equation, let $v = (1+r)^{-1}$ or $x = (1+r)^{-2k}$, solve quadratic/polynomial. Answer in terms of k.
Q2 type — Loan refinancing: (i) Use annuity formula to find L/X (match L =PV of annuity). (ii) Find balance after m payments. Convert interest rate to *effective per period*. New annuity equation for Y. **Key trap:** Lump sum paid immediately, then first instalment one month later (ordinary annuity shifted by 1 month in timeline).
Q3 type — Variable-coupon bond + $R \neq F$: Split into 4 annuity blocks (10 semi-annual payments each at different rates). Write PV equation with P on both sides (since $R = P + 20$), solve linear equation.
Q4 type — Newton–Raphson for bond yield: Set $f(\lambda)$ = PV of cash flows at $\lambda - P_{\text{market}}$. Compute $f(0)$ as a starting check, choose λ_0 near expected answer, iterate 1–2 times for 4 d.p.

Common Mistakes

- Using nominal rate directly in annuity formula without converting
- Forgetting that annuity-due = ordinary annuity $\times(1+r)$
- Coupon = cF/m (not cF) for semi-annual bonds
- Duration: weight by PV, not raw cash flow
- Forward price: uses *risk-free* rate r, not expected return μ
- Put–call parity: discount factor is e^{-rT} (continuous) or $(1+r)^{-T}$ (discrete)